

What the Week Allows

Review coverage, accountability and AI in UK rail design assurance



Perhaps 1 part in 10. The share of a design submission that gets a proper look, as I knew it when I was the reviewer. Section 2 explains the estimate.

BEFORE THE ARGUMENT

A correction to begin with

In February I published a whitepaper called *The Hidden Cost of Engineering Documentation in AEC and Rail*. It counted the hours an engineer spends producing a design assurance form, and it promised that AI would hand most of them back.

I have spent the 8 months since putting that argument in front of the people it was written for: design managers, contractor's engineers, and the project engineers who review what the supply chain sends in.

They were generous about the idea and direct about the numbers. A room of reviewers told me this summer that my hours did not match their week.¹ The paperwork I had been timing is a small part of what fills it.

Small consultancies said something similar from the other side. Most prepare perhaps 1 form a week at their busiest. An hour saved was not what they were asking for. They wanted to know that nothing in the submission was wrong.

So this edition is a rewrite, and its subject has moved. In February I measured how long the paperwork takes to write. The people who sign it kept pointing me to a different measure: how much of it anybody has the time to read.

THE SHORT VERSION

- Design assurance review in rail is sized by the reviewer's week. In my experience a proper look reaches perhaps 10% of what arrives.
- The cost shows up as review rounds and as weeks in the system, far more than as hours at a keyboard.
- General AI tools write fluently and cannot show their evidence, which is what a signed document needs most.
- AI that reads the whole pack and shows its sources could move coverage a long way with the same team. The judgement and the signature stay with the engineer.

01

A comment that asked what the form already said

A friend who works as a Contractor's Responsible Engineer for design shared a Document Review Notice with me recently.

A fair few comments had come back on his submission. Most were the usual ones. One asked for the design life of the system. It was already stated in the Form G.

He raised it. The Designated Project Engineer was honest. He had not had time to read everything, so he had asked what he always asks on that type of job. If the answers looked right, he would pass it.

I have been that reviewer. I spent 2 years as a Designated Project Engineer on a Network Rail renewals portfolio, and I would not fault him. The week leaves room for a proper look at a fraction of what arrives, and experience fills in the rest. Nor would I fault my friend. The evidence was in the pack.

My read is that the gap sits between the 2 of them, in the process. A designer assembles a document thorough enough to be signed. A reviewer with a full portfolio samples it, and falls back on the questions that experience has taught him to ask. Both are doing the job properly.

The document in the middle is written in full and read in part.

That is the cost I missed in February. It looks different from each side of the table, so it is worth sitting on each side in turn.

02

What the week allows

Start on the reviewer's side.

Nobody designed assurance review to cover a tenth of a design. Review in rail is risk-based, and its depth is agreed for the type of work. In practice the depth is also set by something no standard writes down: how many submissions land on 1 engineer in 1 week.

In my own time as a reviewer, a proper look reached perhaps 10% of what arrived.² Experience covered the rest. Which drawing to open first. Which requirement designers tend to miss. Which submission deserves the closer read. That experience is real, and it is what makes a good reviewer. It is also a quiet way of carrying risk.

A submission, as reviewed today



Read properly Carried on experience

Illustrative. The 10% is my own estimate from practice as a reviewer. It is not a measured industry figure, and section 7 asks for yours.

I do not think the estimate is mine alone. This summer a senior engineer described a team working through around 100 review notices and only skimming the surface of them, because the week does not hold a proper review of each. When someone asked whether an AI tool would run a 100% check, the room's own answer was that the checks done today are risk-based.

None of this is a criticism of reviewers. The ones I know would spend any extra hour on a deeper review.

**The 10% is what resource permits.
Nobody would have chosen it.**

The designer, meanwhile, has to write for a reader who may or may not arrive.

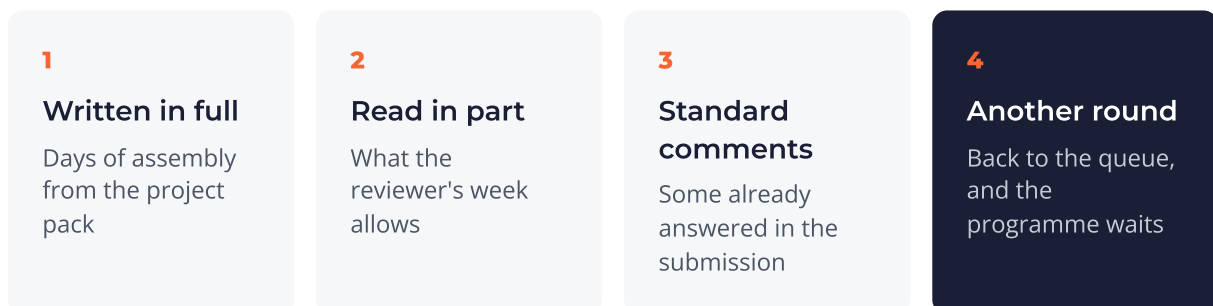
03

The same gap, from the designer's desk

A design manager at a consultancy told me a Form G takes the team a good few days, on schemes where the fee is modest.

An engineer who designs temporary works said that finding the right standards for a submission costs hours each time, and that the form itself tends to get written late on a Friday. Another told me the returned comments take longer to close than the form took to write.

Each of those is a cost of the same gap. The designer cannot know which tenth will be read, so the whole document has to be right. The reviewer cannot read the whole document, so some of the comments that come back are the standard ones. A comment that a phone call would have settled becomes a formal round, and each round sends the submission back to the queue.



The loop as I have seen it from both desks.

A senior engineer on the client side put the consequence to me plainly. The cost that matters is prolongation, with items waiting in the system longer than they should. In renewals, a possession, a hire period and a month of preliminaries can all wait on the same round of comments.

The forms move as well. On 5 September 2026 Network Rail published an addendum to NR/L2/CIV/003, its engineering assurance standard for buildings and civils. All 8 certificate forms were reissued, with compliance from the same day.³ A small change, and an easy one to miss at the point of signature.

So the cost is counted in rounds and in weeks. The obvious remedy is already on most desks.

04

Why a chat window does not close it

Engineers have told me their teams already paste documents into general AI tools, with no agreed method.

The tools write well, and that is the difficulty.

A fluent paragraph with no source looks the same whether it is right or wrong.

A reviewer with a tenth of a week cannot tell which. An engineer described spending much of the time challenging the tool's own output. And a tool that invents a clause reference on a signed certificate is worse than no tool at all.

The regulator's position is short. In its digital safety chapter of 28 September 2026, the Office of Rail and Road expects the risks from AI to be “managed in the same way as any other health and safety risk”, and says that using AI leaves every existing legal duty where it was.⁴ The UK Government's AI Playbook asks that people keep meaningful control at the right stages.⁵

So the signature does not move. Whoever signs a design certificate today signs it tomorrow, whatever drafted the first version. It was the last point raised in that room of reviewers: the comments remain the engineer's comments.

There is a quieter problem too. The model underneath a general tool changes without notice. A study found the same commercial model scoring 84% on a task in March 2023 and 51% on the same task in June.⁶ Somebody has to own the tool after it goes live, and notice when its answers have moved.

Engineers on the client side added 2 conditions that come before any tool. Policy decides which AI may be used at all. And the tool has to reach documents where they live, because exporting a pack and importing the result can cost what the tool saves.

Which leaves a narrower question. What would a tool have to do to sit next to a signed deliverable?

05

What would have to be true

From these conversations I would suggest 6 tests. They apply to any AI tool near a signed deliverable, ours included.

1 It reads the whole pack.

Every document and every page, so nothing depends on which file was opened first.

2 Every statement shows its source.

Document and page, close to hand, so a check takes seconds.

3 It declares what it could not support.

A gap is marked as a gap and referred to the engineer. Nothing is smoothed over.

4 The judgement is asked for, never assumed.

The pack shows what is there today. The design philosophy, the criteria and the chosen solution come from the engineer.

5 The engineer signs.

And the record shows what was checked before the signature.

6 Somebody owns it.

When the model changes or a form is reissued, a named party tests the tool again.

There is a saying I keep coming back to: you can outsource the thinking, but you cannot outsource the understanding. The 6 tests are that saying written down.

A tool that passes them changes what a week can hold.

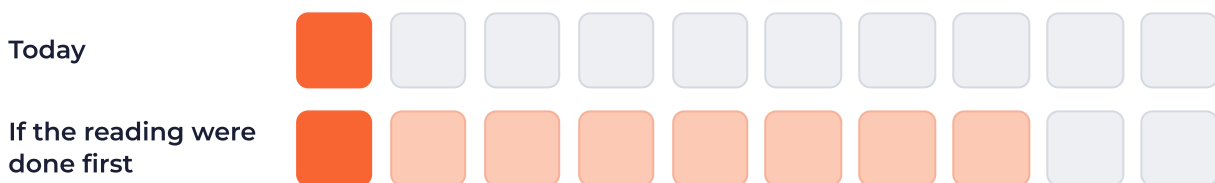
06

If the reading were done first

Go back to the comment about design life.

Suppose the whole submission had been read before the reviewer opened it, and the evidence for the usual questions was already laid out, each item with its page. The design life is there, with the sentence that states it. The reviewer checks it in seconds and spends the hour on the part of the design that needs an engineer.

Perhaps coverage then moves from 10% towards 80%, with the same team in the same week. I hold that number lightly. It is a direction more than a forecast, and nobody has measured it on a live programme yet.



Illustrative. Both bars are my own estimates, the second a direction of travel. Same engineers, same week.

It would not make anyone's week lighter.

The same engineers would be busier, with more of each design in front of them and more decisions to make.

Those are the decisions they trained for. In rail that reads first as a safety argument, because the risk carried quietly in the unread part gets smaller. The same spend on assurance buys more assurance.

The designer gains as well. A submission whose evidence is already set out should draw fewer of the standard comments, and so fewer rounds. Fewer rounds is where the weeks come back.

I should declare an interest at this point.

07

Where Scribtive fits

AutomateX is building Scribtive® OS against those 6 tests. It helps engineers, designers and consultancies prepare PACE assurance deliverables, such as Form G and Form C.

WHAT IT DOES TODAY

- Checks every page you upload.
- Cites document and page in the Word draft where a statement rests on accepted evidence.
- Marks a section for you to complete where it has no accepted evidence.
- Asks the engineer to accept or reject each piece of evidence, and to supply the design judgement.
- Returns the draft in the form's own layout, as a Word file.

WHAT IT DOES NOT DO

- It does not sign anything. Your engineer signs.
- It does not check the design or run a calculation.
- It does not complete the checker's or Network Rail's parts of a form.
- It does not replace an engineer reading their own submission.

It is early. Form G is the first deliverable and Form C is next. It does not yet pass every test on every statement. That is the work of the coming months. We are bringing teams on by demonstration, 1 at a time. The aim is a Form G that arrives close to ready for the engineer to sign and send.

I have not measured the coverage shift on a live programme, and I would rather show you the product than promise a percentage.

If you think 10% is wrong for your team, in either direction, I would like to hear the number you would put on it.

If your team prepares or reviews these deliverables, sign up and I will walk you through a demo.

scribtive.automatex.uk

ABOUT THE AUTHOR



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Bill is the founder of AutomateX, which builds AI for UK rail engineering teams. He has spent 10 years on both sides of the rail delivery table: as a Designated Project Engineer at Network Rail, as an Engineering Manager at Jacobs, leading significant programmes on HS2 as Engineering Lead, and as a director at an SME design consultancy.

He is a Fellow of the Institution of Civil Engineers, Vice Chair of the ICE London Regional Executive Board, and was Young Railway Professional of the Year in 2023.

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SOURCES AND NOTES

1. Conversations with design managers, contractor's engineers and client-side project engineers, February to September 2026, are paraphrased and left unattributed.
2. The 10% and 80% coverage figures are the author's own estimates from practice. They are not measured industry statistics.
3. Network Rail, NR/L2/CIV/003 issue 9 (6 December 2025) and NR/L2/CIV/003/ADD issue 1 (5 September 2026). Cited by reference, issue and date only.
4. Office of Rail and Road, *Digital safety*, strategic risk chapter, 28 September 2026, Artificial Intelligence section (p.18 of the PDF). [orr.gov.uk/digital-safety](https://www.orr.gov.uk/digital-safety). Contains public sector information licensed under the Open Government Licence v3.0.
5. UK Government, *Artificial Intelligence Playbook for the UK Government*, 10 February 2025, Principle 4.
6. L. Chen, M. Zaharia and J. Zou, *How is ChatGPT's behavior changing over time?*, arXiv:2307.09009, 2023. The task was telling prime numbers from composite numbers.

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